TASK 1 – DBMS PROJECT

GROUP 7

GROUP MEMBERS:

Doshi, Pratham Milankumar
Khanna, Rakshit
Lakhani, Vimal Shantibhai
Solanki, Neel Paresh

Neel Solanki

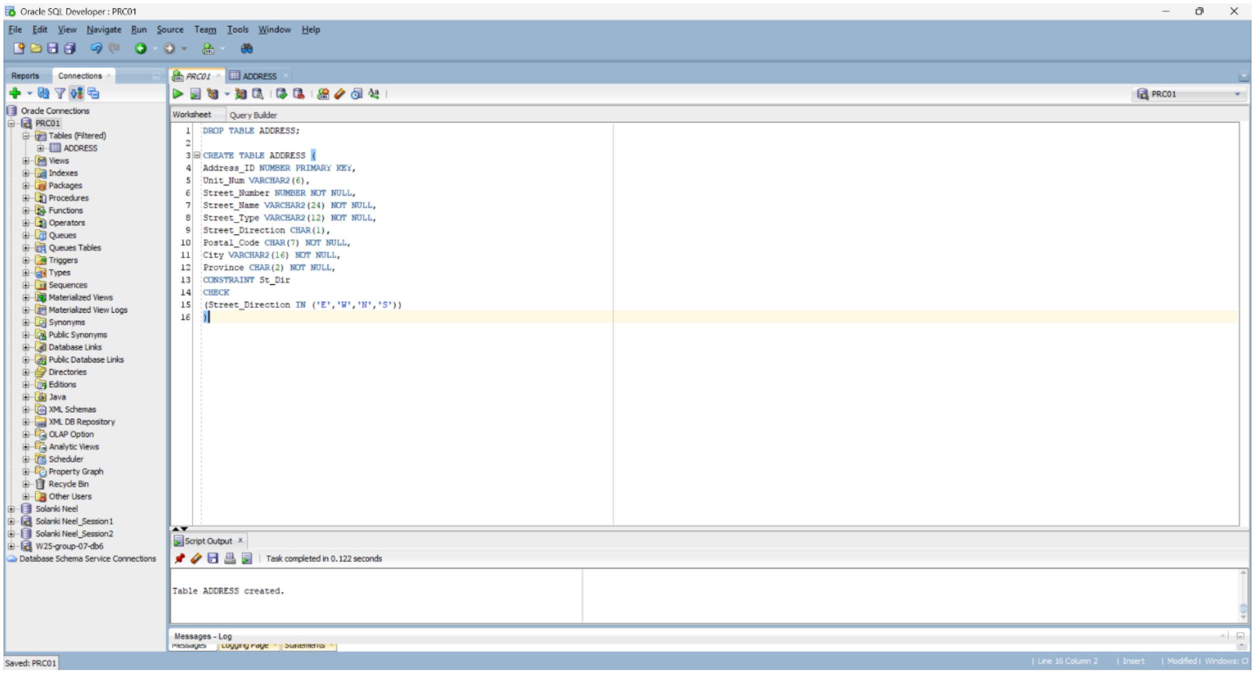
TASK 1

TASK DISTRIBUTION

NEEL PARESH SOLANKI	Designed and implemented the Talend job designs, including metadata connections and data transformation logic.
VIMAL SHANTIBHAI LAKHANI	Set up the Oracle database using SQL Developer and created the address table.
PRATHAM MILANKUMAR DOSHI	Prepared this Project Document and helped in step-by-step implementation
RAKSHIT KHANNA	Provided support during the project by assisting with test runs and helping ensure the overall process went smoothly

Creating Tables and Connections in Oracle SQL Developer for Task 1

1. Creating Address Table



CREATE TABLE ADDRESS (

Address_ID NUMBER PRIMARY KEY,

Unit_Num VARCHAR2(6),

Street_Number NUMBER NOT NULL,

Street_Name VARCHAR2(24) NOT NULL,

Street_Type VARCHAR2(12) NOT NULL,

Street_Direction CHAR(1),

Postal_Code CHAR(7) NOT NULL,

City VARCHAR2(16) NOT NULL,

Province CHAR(2) NOT NULL,

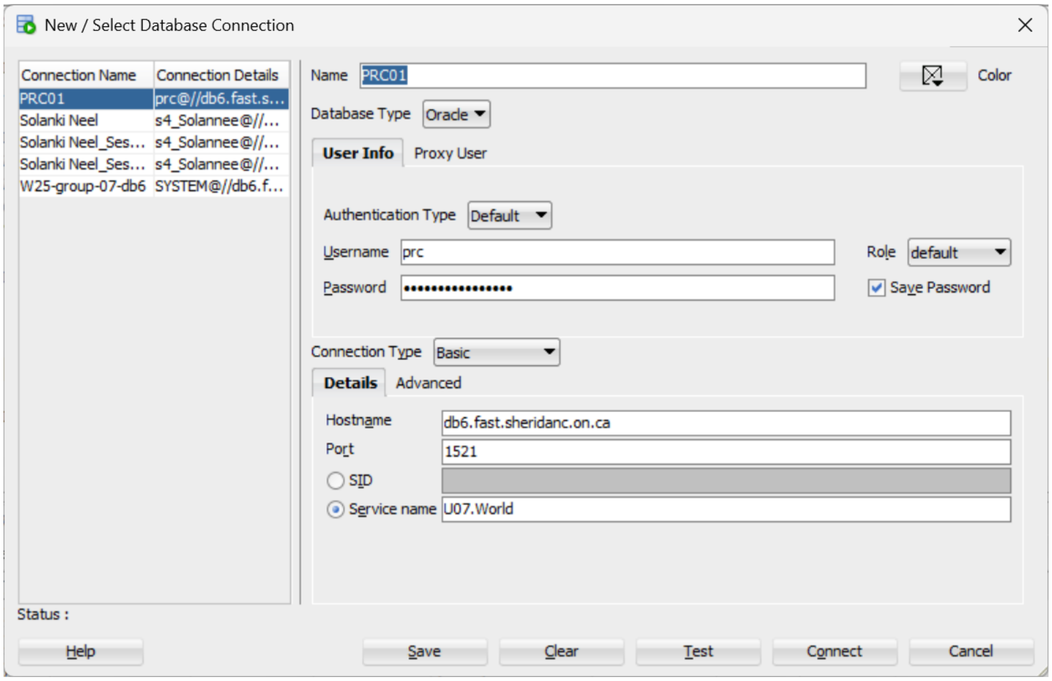
CONSTRAINT St_Dir

CHECK

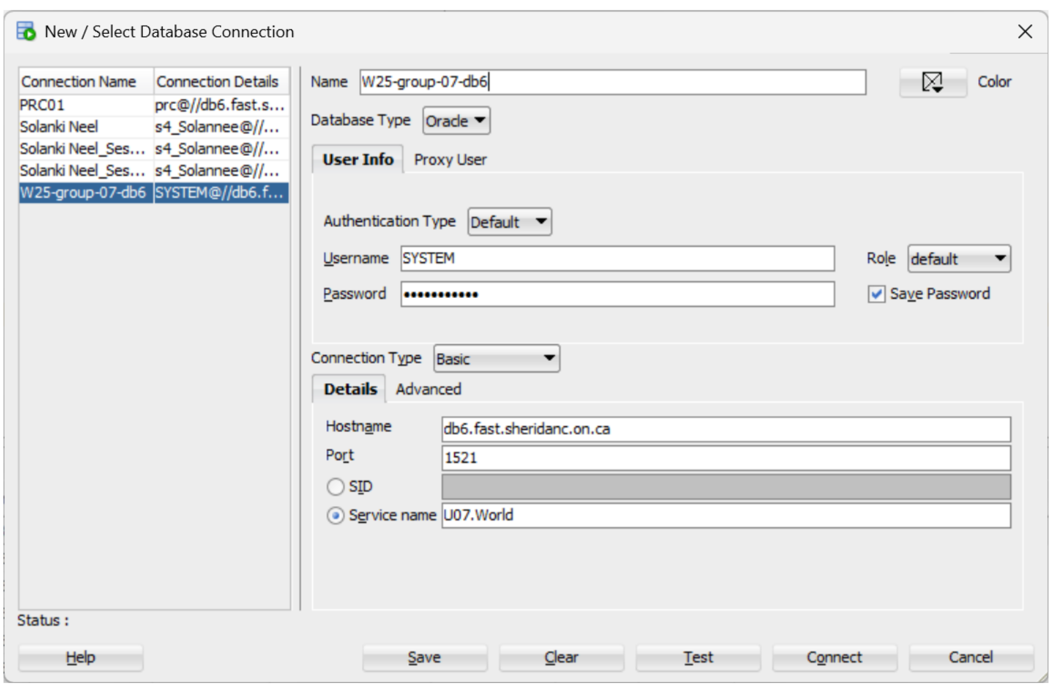
(Street_Direction IN ('E','W','N','S'))

)

2. **Created ORACLE CONNECTION Named PRC01**

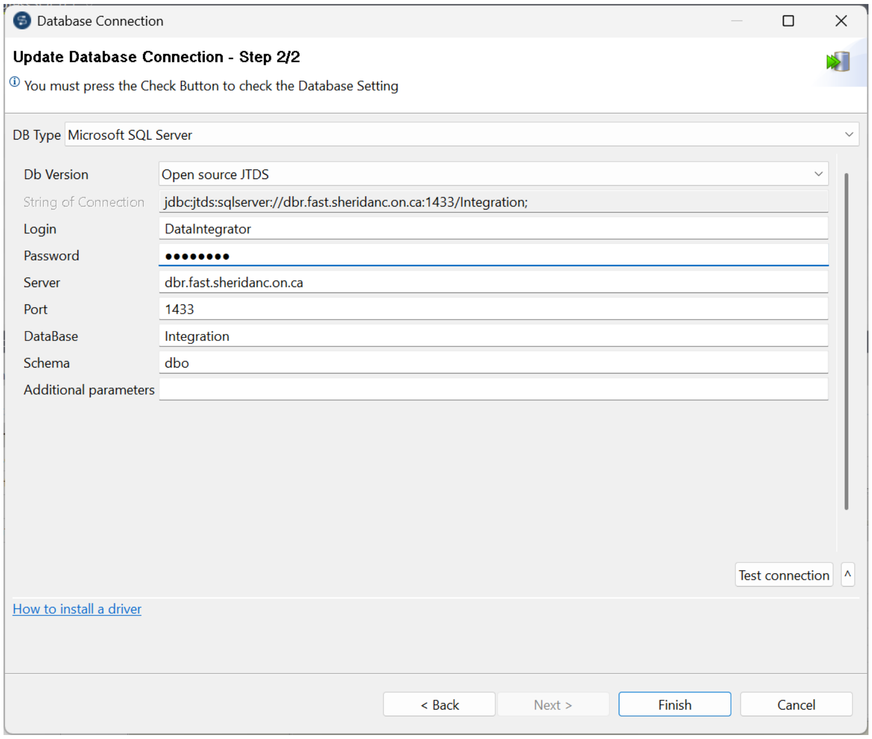


3. **Created ORACLE CONNECTION Named W25-group-07-db6**

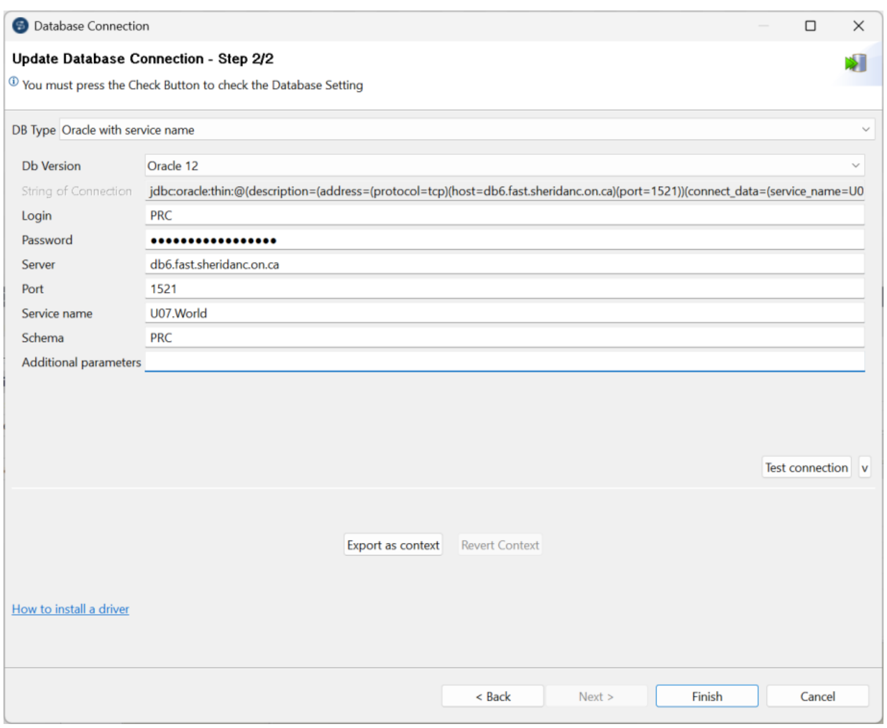


Creating Metadata DB Connection in Talend to connect with Oracle Database

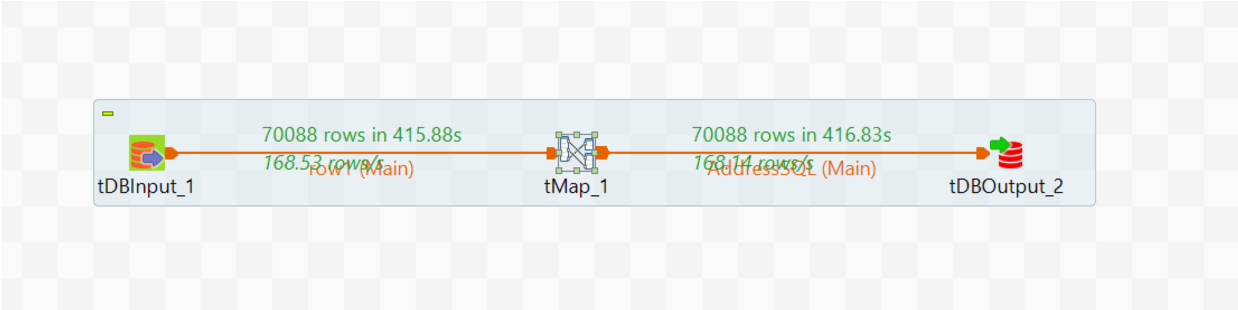
1. Created DB connection in Metadata using **MICROSOFT SQL SERVER** and named as **U07.World**



2. Created DB connection in Metadata using **ORACLE WITH SERVICE NAME** and named as **AddressOracle**.



3. **Job Design Connection**



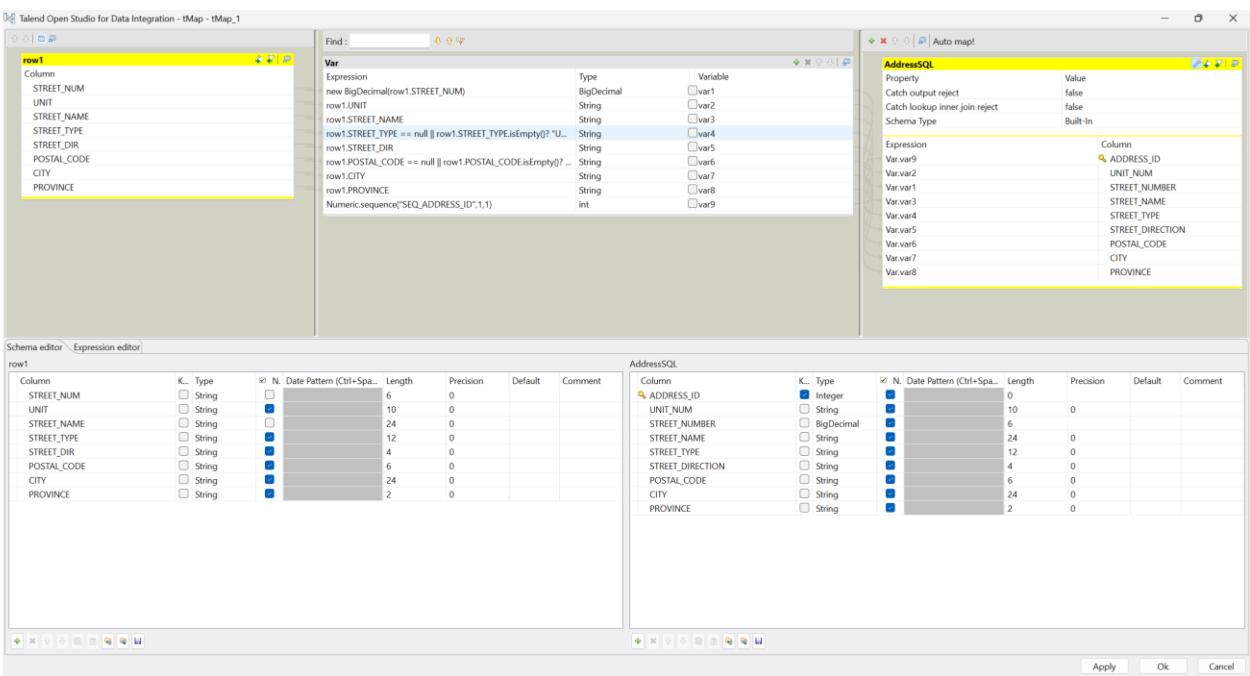
Here In this Connection:

a) **tDBInput_1**: Reads data from a database table (Address Table) and sends 70,088 rows to the next component.

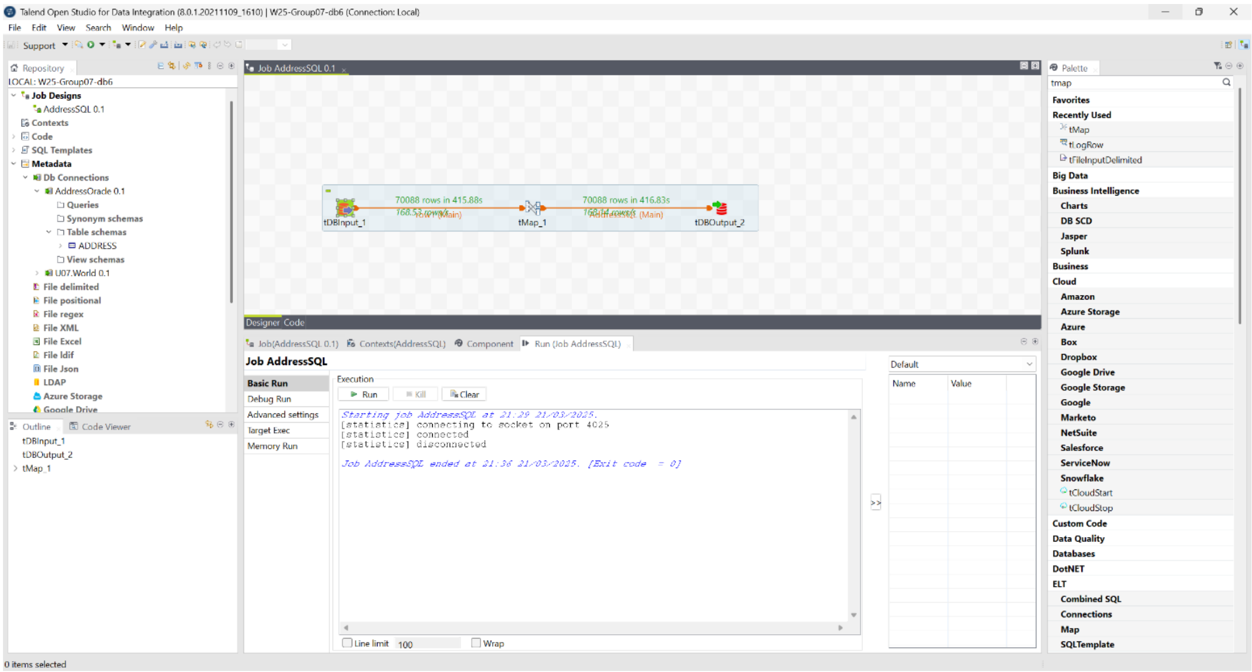
b) **tMap_1**: Transforms and maps the incoming data from tDBInput_1 to the output structure.

c) **tDBOutput_2**: Writes the transformed data into a destination database table.

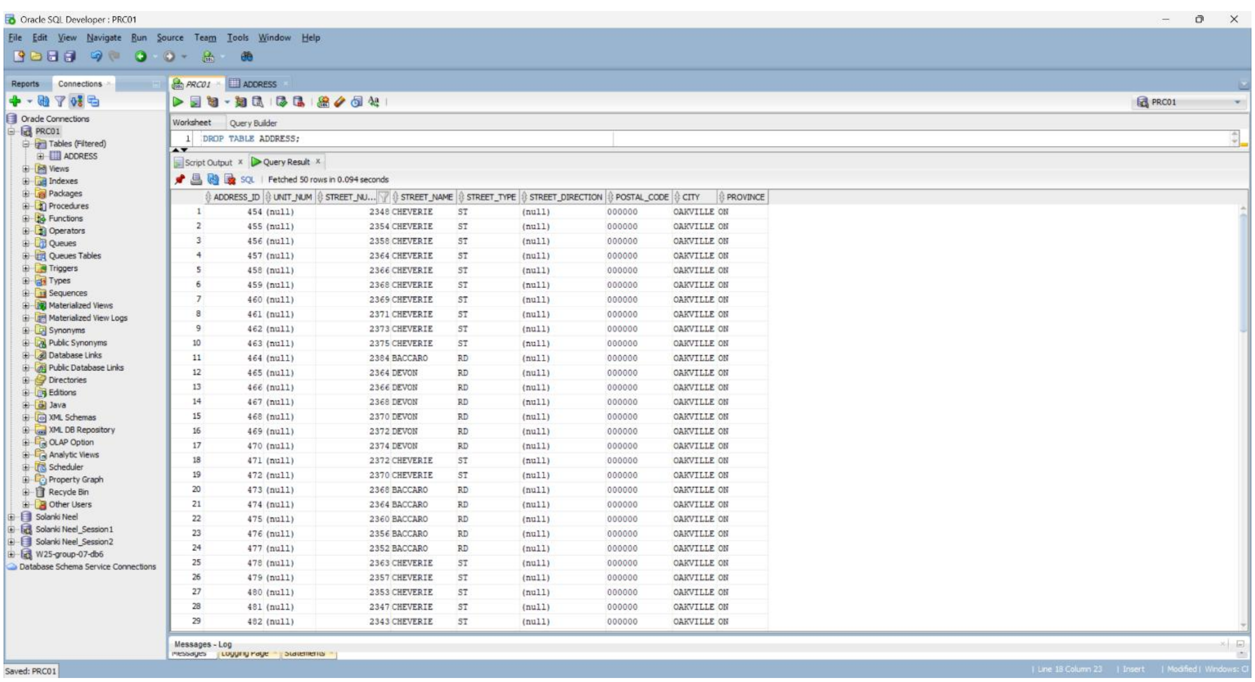
4. **Detailed tMap**



5. Job Design Success for fetching Address Rows

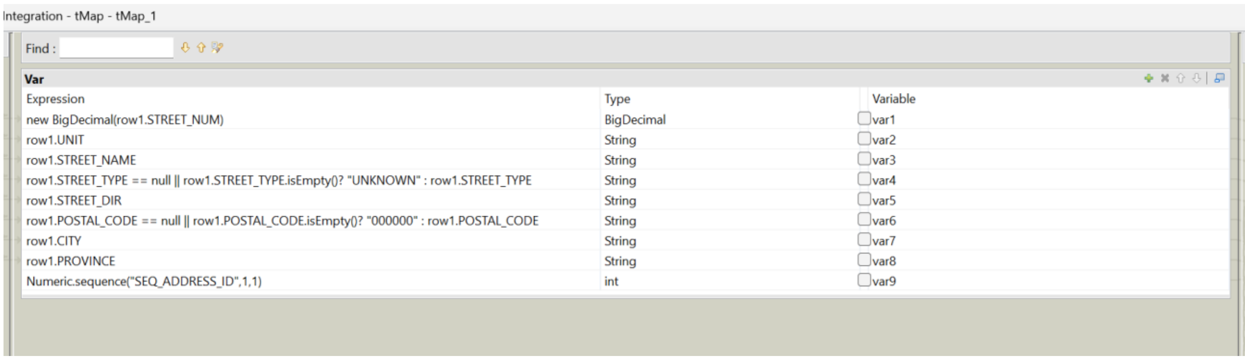


6. Final Address Table with all Values Inserted



	ADDRESS_ID	UNIT_NUM	STREET_NU...	STREET_NAME	STREET_TYPE	STREET_DIRECTION	POSTAL_CODE	CITY	PROVINCE
1	454	(null)	2348	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
2	455	(null)	2354	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
3	456	(null)	2358	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
4	457	(null)	2364	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
5	458	(null)	2366	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
6	459	(null)	2368	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
7	460	(null)	2369	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
8	461	(null)	2371	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
9	462	(null)	2373	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
10	463	(null)	2375	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
11	464	(null)	2384	BACCARO	RD	(null)	000000	OAKVILLE	ON
12	465	(null)	2364	DEVON	RD	(null)	000000	OAKVILLE	ON
13	466	(null)	2366	DEVON	RD	(null)	000000	OAKVILLE	ON
14	467	(null)	2368	DEVON	RD	(null)	000000	OAKVILLE	ON
15	468	(null)	2370	DEVON	RD	(null)	000000	OAKVILLE	ON
16	469	(null)	2372	DEVON	RD	(null)	000000	OAKVILLE	ON
17	470	(null)	2374	DEVON	RD	(null)	000000	OAKVILLE	ON
18	471	(null)	2372	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
19	472	(null)	2370	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
20	473	(null)	2368	BACCARO	RD	(null)	000000	OAKVILLE	ON
21	474	(null)	2364	BACCARO	RD	(null)	000000	OAKVILLE	ON
22	475	(null)	2360	BACCARO	RD	(null)	000000	OAKVILLE	ON
23	476	(null)	2356	BACCARO	RD	(null)	000000	OAKVILLE	ON
24	477	(null)	2352	BACCARO	RD	(null)	000000	OAKVILLE	ON
25	478	(null)	2363	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
26	479	(null)	2357	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
27	480	(null)	2353	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
28	481	(null)	2347	CHEVERIE	ST	(null)	000000	OAKVILLE	ON
29	482	(null)	2343	CHEVERIE	ST	(null)	000000	OAKVILLE	ON

7. Changes Made in TMap to solve the error I was getting before, Inserted some Query.



Integration - tMap - tMap_1

Find :

Var

Expression	Type	Variable
new BigDecimal(row1.STREET_NUM)	BigDecimal	var1
row1.UNIT	String	var2
row1.STREET_NAME	String	var3
row1.STREET_TYPE == null || row1.STREET_TYPE.isEmpty()? "UNKNOWN" : row1.STREET_TYPE	String	var4
row1.STREET_DIR	String	var5
row1.POSTAL_CODE == null || row1.POSTAL_CODE.isEmpty()? "000000" : row1.POSTAL_CODE	String	var6
row1.CITY	String	var7
row1.PROVINCE	String	var8
Numeric.sequence("SEQ_ADDRESS_ID",1,1)	int	var9

1. **POSTAL_CODE** = row1.POSTAL_CODE == **null** || row1.POSTAL_CODE.isEmpty()? "000000" : row1.POSTAL_CODE

2. **ADDRESS_ID** = Numeric.sequence("SEQ_ADDRESS_ID",1,1)

3. **STREET_TYPE** = row1.STREET_TYPE == null || row1.STREET_TYPE.isEmpty()? "UNKNOWN" : row1.STREET_TYPE